

Getting Started in WRDS - CRSP Data Extraction with Python and SQL

QF600 Asset Pricing

25 Sep 2026

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Senior Librarian, Instruction & Learning Services

What is WRDS?



Wharton
UNIVERSITY of PENNSYLVANIA

wrds WHARTON RESEARCH
DATA SERVICES

- **Wharton Research Data Services (WRDS)** provides access to over 350 terabytes of data across multiple disciplines including Accounting, Banking, Economics, ESG, Finance, Healthcare, Insurance, Marketing, and Statistics.
- It hosts a wide range of datasets from over 70 vendors, e.g. **Center for Research in Security Prices (CRSP)**.
- Data delivery options include a **web-based query form** and programming using **Python**, SAS, R, Stata, and MATLAB.

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[WRDS Request for Help](#)

Part I

Accessing WRDS via SMU Libraries

Access WRDS via SMU Libraries



Libraries

Menu

Go to: library.smu.edu.sg

Everything

Search for books, articles, databases and more



Shortcuts:

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JOURNALS

NEWSPAPERS

AI-LITERACY WORKSHOPS

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INK

OPEN RESOURCES



Next Day

MONDAY, AUGUST 17, 2026

Li Ka Shing Library	8:00am – 11:59pm
Learning Commons	24/7
Kwa Geok Choo Law Library	8:00am – 11:59pm
Learning Commons	24/7
"Ask a Librarian" Chat	10:00am – 5:00pm

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Ask a Librarian

CURRENT OCCUPANCY

MON, 17 AUG 2026Li Ka Shing Library
588 / 1800Kwa Geok Choo Law Library
127 / 500Chat with SMU
Libraries AI
chatbot[View Occupancy Details](#)**Get It For Me »**Request to obtain resources not in
the collection through interlibrary**Book A Computer »**Book a computer in the Investment
and Data Studio**Library FAQ »**

Quick answers to your questions

Register for your individual account with WRDS

A-Z Databases

Find the best library databases for your research.

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13 Databases

All A B C D E F G H I J K L M N O P Q R S T U V **W** X Y Z #

W

[Wharton Research Data Services \(WRDS\)](#)

WRDS provides access to over 350 terabytes of data across multiple disciplines including Accounting, Banking, Economics, ESG, Finance, Healthcare, Insurance, Marketing, and Statistics. Data delivery options include a web query method, the WRDS Cloud for executing research and strategy development, and the WRDS client server using PCSAS, Matlab, and R.

[More](#)

Registration is required for access to WRDS.

[Register for your individual account](#) or [obtain a Day Pass access](#) with your SMU email.

Register for your individual account with WRDS
(Remember to use your SMU email address)

[Analytics](#)[Data](#)[Classroom](#)[About](#) ▾[Videos](#)[Demo](#)[LOGIN](#) [REGISTER](#) [PASSWORD RESET](#) [ACCOUNT TRANSFER](#) [CONTACT](#)[Home](#)

Register for WRDS

Username

Please choose a username.

Required. Usernames must be 4 to 15 characters long; begin with a letter; contain only lowercase ASCII letters, numbers, and/or underscores; and must not begin with pg_.

Password

Please choose a password.

- Your password must contain at least 12 characters.
- Your password can't be a commonly used password.
- Your password can't be entirely numeric.
- Passwords may contain only ASCII letters, numbers, spaces, and punctuation characters: `~!@#\$%^&*()-=_+[]\{}|;':",./<>?

Password confirmation

Please choose a password.

Enter the same password as before, for verification.

First name

Last name

Login your individual account (2FA required)

LOGIN REGISTER PASSWORD RESET ACCOUNT

Wharton UNIVERSITY of PENNSYLVANIA wrds WHARTON RESEARCH DATA SERVICES

smustf1926

Login

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wrds

The Global Standard for Business Research

Enabling impactful scholarly research

01:39

vimeo

Make sure that your mobile device is not connected to any VPN while receiving the 2FA notification from WRDS!

WRDS Enables Impactful Research

GLOBAL DATA

RESEARCH ANALYTICS

ACADEMIC SUPPORT

For **30+ years**, Wharton Research Data Services (WRDS) has supported users with targeted solutions that support **research**, reinforce **learning**, and enable **discovery**. **WRDS** advances comprehensive thought leadership by allowing users to **analyze** complex data and develop academic research.

Access CRSP through WRDS



wrds

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DATA SERVICES



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Data Dashboard »

» Compustat

» IR

» SEC Analytics Suite by WRDS

» Contributed Data at WRDS **33**

» ISS ESG **60**

» SEC Order Execution **1**

» CRSP **346**

» ISSM **2**

» Subsidiary Data by WRDS **1**

» Efficient Frontier by WRDS **1**

» LSEG FTSE Russell **3**

» Thomson / Refinitiv **11**

» Environmental Suite by WRDS **4**

» LSEG IBES **114**

» TRACE **32**

» ETF Global **4**

» Macro Finance Society **27**

» US Patents by WRDS **3**

» European Short Data by WRDS **1**

» MFLINKS **3**

» Venture Capital by WRDS **6**

» Event Study by WRDS **10**

» MSRB **1**

» World Indices by WRDS **3**

The **Center for Research in Security Prices, LLC (CRSP)** maintains the most comprehensive collection of security price, return, and volume data for the **US equity market**, e.g. NYSE, AMEX and NASDAQ stock markets

</> Coding Tool Kit

SAS Studio

About

R Studio

About

JupyterHub

About

Examples

WRDS News

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AUG

Downtime Scheduled for
August 26, 2026

August 11th, 2026

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AUG

Research Webinar | I/B/E/S
Unpacked | August 20, 2026

August 10th, 2026

Access CRSP datasets

CRSP

Founded at the University of Chicago, CRSP has transformed the world of finance. More than 65 years ago, CRSP developed the first market database that allowed investors to measure historic rates of return for U.S. stocks, broadening the appeal of equity ownership. More than 15 years ago, CRSP launched investable indexes that have become trusted by investors globally, including managers of the world's largest mutual fund.



[More About This Vendor](#) »

Legacy Data Has Moved

The SIZ version of CRSP Stock and Index data has moved to the Legacy section below. Please see [this note](#) for details.

Related Topics

- [Bonds and Fixed Income](#)
- [Event Study](#)
- [Indices and Factors](#)
- [Institutional Ownership Research](#)
- [Mutual Fund / Hedge Fund / ETF Returns](#)
- [Ownership](#)
- [Portfolio Construction and Market Anomalies](#)
- [REITs](#)
- [Stock Prices](#)
- [North America](#)

Annual Update

Databases in this section are updated once each year, in early February. Update schedules should not be confused with end-of-day, end-of-month, or end-of-quarter data such as stock prices.

» [Stock - Version 2 \(CIZ\)](#)

9

» [Index - Version 2 \(CIZ\)](#)

10

» [Treasuries](#)

11

Use this dataset to extract updated stock level data

Knowledge Base

» [Stock v1 \(SIZ\)](#)

» [Index and Deciles v1 \(SIZ\)](#)

» [ETF Identification](#)

Manuals and Overviews

» [WRDS Overview of Legacy CRSP U.S. Stock Database](#)

» [WRDS Overview of CRSP/COMPUSTAT Merged \(CCM\)](#)

» [Stocks and Indices](#)

Additional Tools

Access CRSP Daily Stock data

Stock - Version 2 (CIZ)

Founded at the University of Chicago, CRSP has transformed the world of finance. More than 65 years ago, CRSP developed the first market database that allowed investors to measure historic rates of return for U.S. stocks, broadening the appeal of equity ownership. More than 15 years ago, CRSP launched investable indexes that have become trusted by investors globally, including managers of the world's largest mutual fund.

[» Daily Stock File](#)

Click to enter Daily Stock query form

[» Delisting Information](#)[» Distribution](#)[» Monthly Stock File](#)[» Monthly Stock Market Indexes](#)[» Names](#)[» Share Outstanding](#)[» Stock Header Information](#)[More About This Vendor »](#)

Knowledge Base »

[» Cumulative Adjustment Factors in Stock v2](#)

Manuals and Overviews »

Additional Tools



Merging CRSP and
Compustat Data 1: Using
Historical CUSIPs

[More →](#)

CRSP Basics

[More →](#)

CRSP Daily Stock File query form

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Directory

Daily Stock File



- Query Form
- Variable Descriptions
- Manuals and Overviews
- Knowledge Base
- Data Preview

Step-by-step query form

Stock Daily Stock File contains the prices, capital structure, and other information and is accessed by the CRSP unique stock identifier, DlyCalDt.

Step 1: Choose your date range.

Daily Calendar Date (DlyCalDt)

2020-01-01

to

2024-12-31

Variable Descriptions: Quick reference on the variables

Manuals and Overviews: [\[Important\] CRSP US Stock & Indices Guide](#)

Knowledge Base: Useful FAQs

Data Preview: A preview of the daily stock data

Step 2: Apply your company codes.

What format are your company codes?

Autocomplete

☒ (permno) PERMNO

☐ (permco) PERMCO

☐ (ticker) Ticker

☐ Full header CUSIP

☐ (sized) Sin Code

☐ Header CUSIP

Annual Update

Stock - Version 2 (CIZ) (hide)

>> Daily Stock File

>> Daily Stock Market Indexes

>> Delisting Information

CRSP Daily Stock Variable Descriptions

Daily Stock File

Query Form	Variable Descriptions	Manuals and Overviews	Knowledge Base	Data Preview
------------	-----------------------	-----------------------	----------------	--------------

Product:	crsp_a_stock
Library:	crspa
File:	wrds_dsfv2_query
Date Range:	(dlycaldt) Daily Calendar Date: 1925-12-31 - 2025-12-31
Last Updated On:	2026-07-16
Update Frequency:	Annually

Variable Reference

Variable Name	Type	Description
permno	Int	(permno) PERMNO
secinfostartdt	Date	(secinfostartdt) Security Information Start Date
secinfoenddt	Date	(secinfoenddt) Security Information End Date
securitybegdt	Date	(securitybegdt) Begin Date of Stock Data



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[Stock - Version 2 \(CIZ\)](#) (hide)

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» [Daily Stock Market Indexes](#)

» [Delisting Information](#)

» [Distribution](#)

» [Monthly Stock File](#)

» [Monthly Stock Market Indexes](#)

CRSP US Stock & Indices Guide

Daily Stock File

Query Form	Variable Descriptions	Manuals and Overviews	Knowledge Base	Data Preview
U.S. Stock and Indices Guide (...)		SIZ to CIZ: CRSP Executive s...	» SIZ to CIZ: WRDS Sample Code	
CRSP Release Notes		» SIZ to CIZ: WRDS Overview of Index	CRSP Metadata Tables Guide ...	
CRSP CIZ Calculations and In...		» SIZ to CIZ: WRDS Transition FAQ - Stock	» SIZ to CIZ: Recreate Legacy MSE Style Tables	
» SIZ to CIZ: WRDS Overview of Stock		SIZ to CIZ: CRSP Cross Refer...		

Manuals and Overviews: [\[Important\] CRSP US Stock & Indices Guide](#)

[CRSP 10 Year U.S. Stock Dat...](#)

[About CRSP Select Series](#)

[CRSPMI 2013 Historical Indexes](#)

[CRSP Flat File formats and no...](#)

[CRSPMI 2013 Historical Mark...](#)

[Research Indexes \(Indices\) G...](#)



**CENTER FOR RESEARCH
IN SECURITY PRICES**

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[Stock - Version 2 \(CIZ\) \(hide\)](#)

[» Daily Stock File](#)

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[» Distribution](#)

[» Monthly Stock File](#)

[» Monthly Stock Market Indexes](#)

Query Form	Variable Descriptions	Manuals and Overviews	Knowledge Base	Data Preview
------------	-----------------------	-----------------------	----------------	--------------

This preview is for the `crsp_a_stock.wrds_dsfv2_query` table.
Query Forms often combine multiple tables and may contain additional columns/variables.

Permno	Secinfostrt	Secinfoend	Securitybegdt	Securityenddt	Securityhdrflg	Hdrcusip
10026	2019-08-05	2021-04-26	1986-02-04	2024-12-31	N	46603
10028	2019-12-18	2020-02-23	1986-02-05	2024-12-31	N	29402
10032	2019-09-12	2023-09-28	1986-02-05	2024-12-31	N	
10044	2017-12-12	2021-02-22	1986-02-11	2024-12-31	N	
10051	2018-09-12	2020-02-13	1986-02-14	2022-11-03	N	
10065	2019-08-09	2023-09-28	1925-12-31	2024-12-31	N	
10066	2008-01-22	2021-03-28	1983-08-25	2024-12-31	N	
10104	2013-07-15	2023-06-26	1986-03-12	2024-12-31	N	

For example, definitions of the variables (security identifiers, ranges, descriptors) are available on p.13 of the guide.

StkSecurityInfoHdr & StkSecurityInfoHist – STOCK SECURITY INFORMATION HEADER & HISTORY

Tables below include security identifiers, ranges, descriptors. Header includes the most recent values for each item. History includes a compilation of item values over the history of the security.

Item Name	Item Description	Item Category	Item Definition
PERMNO	PERMNO	Key	CRSP unique permanent security level identifier
SecInfoStartDt	Security Information Start Date	Date	The start date of the data row that corresponds with the security End Date in the same row
SecInfoEndDt	Security Information End Date	Date	The end date of the data row that corresponds with the security Start Date in the same row
SecurityBegDt	Begin Date of Stock Data	Date	Starting date for security daily time series data in YYYY-MM-DD format
SecurityEndDt	End Date of Stock Data	Date	Ending date for security daily time series data in YYYY-MM-DD format
SecurityHdrFlg	Security Header Flag	Flag	Y/N flag – uniformly Yes for the StkSecurityInfoHdr file
HdrCUSIP	Header CUSIP	Id	Most recent CUSIP on file or dummy CUSIP
HdrCUSIP9	Header CUSIP 9	Id	Most recent CUSIP on file or dummy CUSIP. Same as the Header CUSIP with an additional check digit
CUSIP	CUSIP	Id	CUSIP without check digit
CUSIP9	CUSIP9	Id	CUSIP with check digit
PrimaryExch	Primary Exchange	Flag	Identifies the exchange on which the security has its primary listing
ConditionalType	Conditional Type	Flag	Describes the status of a security
ExchangeTier	Exchange Tier	Flag	Identifies the NASDAQ Market Tier

This preview is for the `crsp_a_stock.wrds_dsfv2_query` table.
Query Forms often combine multiple tables and may contain additional columns/variables.

Permno ◊	Secinfostrt	Secinfoend	Securitybegdt	Securityenddt	Securityhdrflg	HdrCUSIP	Search Filters...
10026	2019-08-05	2021-04-26	1986-02-04	2024-12-31	N	46603	> Permno
10028	2019-12-18	2020-02-23	1986-02-05	2024-12-31	N	29402	> Permno
10032	2019-09-12	2023-09-28	1986-02-05	2024-12-31	N		
10044	2017-12-12	2021-02-22	1986-02-11	2024-12-31	N		
10051	2018-09-12	2020-02-13	1986-02-14	2022-10-03	N		
10065	2019-08-09	2023-09-28	1925-12-31	2024-12-31	N		
10066	2008-01-22	2021-03-28	1983-08-25	2024-12-31	N		
10104	2013-07-15	2023-06-26	1986-03-12	2024-12-31	N		

StkSecurityInfoHdr & StkSecurityInfoHist – STOCK SECURITY INFORMATION HEADER & HISTORY

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SecInfoEndDt	Security Information End Date	Date	The end date of the data row that corresponds with the security Start Date in the same row
SecurityBegDt	Begin Date of Stock Data	Date	Starting date for security daily time series data in YYYY-MM-DD format
SecurityEndDt	End Date of Stock Data	Date	Ending date for security daily time series data in YYYY-MM-DD format
SecurityHdrFlg	Security Header Flag	Flag	Y/N flag – uniformly Yes for the StkSecurityInfoHdr file
HdrCUSIP	Header CUSIP	Id	Most recent CUSIP on file or dummy CUSIP
HdrCUSIP9	Header CUSIP 9	Id	Most recent CUSIP on file or dummy CUSIP. Same as the Header CUSIP with an additional check digit
CUSIP	CUSIP	Id	CUSIP without check digit
CUSIP9	CUSIP9	Id	CUSIP with check digit
PrimaryExch	Primary Exchange	Flag	Identifies the exchange on which the security has its primary listing
ConditionalType	Conditional Type	Flag	Describes the status of a security
ExchangeTier	Exchange Tier	Flag	Identifies the NASDAQ Market Tier

PERMNO: CRSP unique, permanent security/stock level identifier



*“Daily stock data for Apple for the
year 2022.”*

Part III

Querying WRDS Data using Python

Coding for Data Comprehension / EDA



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Python

How to access WRDS data through Python: from your computer or on the WRDS Cloud

[» Python: From the Web \(JupyterHub\)](#)[» PYTHON: On the WRDS Cloud](#)[» PYTHON: From Your Computer
\(Jupyter/Spyder\)](#)[» Reducing Memory Usage](#)

Wharton Research Data Services

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[Conference Calendar](#)
[Impactful Research](#)

MQF.ipynb (auto-G) - JupyterLab

← → ↻ wrds-jupyter.wharton.upenn.edu/user/ltee11/lab/workspaces/auto-G/tree/MQF.ipynb 🔍 🔗 ☆ ⚙️ 👤 ⋮

File Edit View Run Kernel Tabs Settings Help

+ + ↻

Filter files by name 🔍

/ web_query_output /

Name ▲	Last Modified
📄 mstrxxciefzki6zc_csv.zip	an hour ago

MQF.ipynb × +

📄 + ✂️ 📄 📋 ▶️ ■ ↻ ⏏️ Code ▾ ⚙️ Python 3 (ipykernel) ○

```
[129]: import wrds
db = wrds.Connection()
lib = db.list_libraries()
lib
#sorted(lib)

Loading library list...
Done

[129]: ['aha_sample',
      'ahasamp',
      'audit',
      'audit_audit_comp',
      'audit_common',
      'auditsmp',
      'auditsmp_all',
      'block',
      'block_all',
      'boardex_trial',
      'boardsmp',
      'bvd',
      'bvd_orbis_large',
      'bvd_orbis_medium',
      'bvd_orbis_small',
```

Web:

CRSP > WRDS Update Freq > Query Form.Cluster > Query Form > Data Item

Python:

CRSP > Product/**Library** > **Table** > Data Item

Stock – Version 2 (CIZ)’s Daily Stock File

crsp_a_stock.wrds_dsfv2_query (Version 2)

Daily Stock File

[Query Form](#)[Variable Descriptions](#)[Manuals and Overviews](#)[Knowledge Base](#)[Data Preview](#)

Changes to CRSP Data

New versions of CRSP Stock and CRSP Index have been released with changes to table and variable names. Please see [this note](#) for details.

Posted November 22nd, 2024

Don't show this again ☐

Stock Daily Security Data contains the prices, capitalization, returns, volumes, and other information and is accessed by the CRSP unique security identifier, PERMNO and the daily calendar date, DlyCalDt.

Step 1: Choose your date range.

Daily Calendar Date (DlyCalDt)

to

Step 2: Apply your company codes.

What format are your company codes?

Autocomplete



(permno) PERMNO



(permco) PERMCO



(ticker) Ticker



Full header CUSIP



(siccd) Sic Code



Header

CUSIP



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[CRSP](#) »

[Annual Update](#) »

[Stock - Version 2 \(CIZ\)](#)

(hide)

» [Daily Stock File](#)

» [Daily Stock Market Indexes](#)

» [Delisting Information](#)

» [Distribution](#)

» [Monthly Stock File](#)

» [Monthly Stock Market Indexes](#)

» [Names](#)

» [Share Outstanding](#)

Daily Stock File

Query Form	Variable Descriptions	Manuals and Overviews	Knowledge Base	Data Preview
Product:	crsp_a_stock			
Library:	crspa			
File:	wrds_dsfv2_query			
Date Range:	(dlycaldt) Daily Calendar Date: 1925-12-31 - 2025-12-31			
Last Updated On:	2026-07-16			
Update Frequency:	Annually			

Variable Reference

Variable Name	Type	Description
permno	Int	(permno) PERMNO
secinfostartdt	Date	(secinfostartdt) Security Information Start Date
secinfoenddt	Date	(secinfoenddt) Security Information End Date
securitybegdt	Date	(securitybegdt) Begin Date of Stock Data



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[Stock - Version 2 \(CIZ\)](#) (hide)

» [Daily Stock File](#)

» [Daily Stock Market Indexes](#)

» [Delisting Information](#)

» [Distribution](#)

» [Monthly Stock File](#)

Product: CRSP Stock (Annual) (crsp_a_stock)

Long Description:	None
Last Updated:	Feb. 8, 2025, 3:31 p.m.
Update Frequency:	Annually
Postgres Schema:	crsp_a_stock
Library:	crsp
Path(s) on WRDS Cloud:	/wrds/crsp/sasdata/a_stock/*.*, /wrds/crsp/sasdata/a_stock_v2/*.*, /wrds/crsp/seqdata/sd/*.*, /wrds/crsp/seqdata/sm/*.*, /wrds/crsp/seqdata/ix/*.*
Date Range:	1923-01-01 - 2099-12-31
Total Size:	175.13 GiB
Sentinel Dates:	



Contact

Address:
University of Chicago Booth School of Business
500 W Madison St, Ste 2900
Chicago, IL
United States

📞 312-263-6400
✉ Support@crsp.org
🌐 <http://crsp.org/>

Tables:

Table	Description	Query Form(s)
wrds_dsfv2_query	Daily Stock File query created by WRDS	Daily Stock File
wrds_monthlyindexret_query	Monthly Stock Market Index Return query created by WRDS	Monthly Stock Market Indexes
wrds_msfv2_query	Monthly Stock File query created by WRDS	Monthly Stock File

smu-wrds.python Public

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Code

<https://github.com/hwee-ming/smu-wrds.python>

hwee-ming Rename 26_0925_qf600.ipynb.ipynb to 26_0925_qf600.ipynb

b97454b · last week 5 Commits

26_0907_mqf_brownbag_01.ipynb

26_0914_qf209.ipynb

26_0925_qf600.ipynb

README

Files

main

Go to file

26_0907_mqf_brownbag_01.ipynb

26_0914_qf209.ipynb

26_0925_qf600.ipynb

smu-wrds.python / 26_0925_qf600.ipynb

hwee-ming Rename 26_0925_qf600.ipynb.ipynb to 26_0925_qf600.ipynb

b97454b · last week History

Preview

Code

Blame

1915 lines (1915 loc) · 1.24 MB

Raw

Copy

Download

Edit

More

Open in Colab

```
In [1]: pip install wrds

Requirement already satisfied: wrds in /usr/local/lib/python3.13/dist-packages (3.5.0)
Requirement already satisfied: ipython<9.8.0 in /usr/local/lib/python3.13/dist-packages (from wrds) (7.34.0)
Requirement already satisfied: pandas<2.3,>=2.2 in /usr/local/lib/python3.13/dist-packages (from wrds) (2.2.3)
Requirement already satisfied: psycopy2-binary<2.10,>=2.9 in /usr/local/lib/python3.13/dist-packages (from wrds) (2.9.13)
Requirement already satisfied: sqlalchemy<2.1,>=2 in /usr/local/lib/python3.13/dist-packages (from wrds) (2.0.52)
Requirement already satisfied: setuptools>=18.5 in /usr/local/lib/python3.13/dist-packages (from ipython<9.8.0->wrds) (80.10.2)
Requirement already satisfied: jedi>=0.16 in /usr/local/lib/python3.13/dist-packages (from ipython<9.8.0->wrds) (0.20.0)
Requirement already satisfied: decorator in /usr/local/lib/python3.13/dist-packages (from ipython<9.8.0->wrds) (4.4.2)
Requirement already satisfied: pickleshare in /usr/local/lib/python3.13/dist-packages (from ipython<9.8.0->wrds) (0.7.5)
Requirement already satisfied: traitlets>=4.2 in /usr/local/lib/python3.13/dist-packages (from ipython<9.8.0->wrds) (5.7.1)
Requirement already satisfied: prompt-toolkit!=3.0.0,!<3.1.0,>=2.0.0 in /usr/local/lib/python3.13/dist-packages (from ipython<9.8.0->wrds) (3.0.53)
```

Using Python on WRDS Platform



Learn how to use Python on the WRDS platform, including:

- Establishing a connection to the WRDS server
- Using the basic functions provided by the WRDS API

Sample Code

Using Python on WRDS Platform

-- Connection and Basic Function

- Qingyi (Freda) Song Drechsler, Ph.D.
- August 2020

0. Setup WRDS Python API

- To be done before running this notebook
- <https://pypi.org/project/wrds/>
- run "pip install wrds"
 - Mac: in terminal
 - PC: in Anaconda prompt
- Spyder or Jupyter

1. Import pre installed WRDS package

```
In [ ]: import wrds
```

2. Establish connection with WRDS server

- log in using your WRDS username and password
- set up a pgpass file to store the info

```
In [ ]: conn = wrds.Connection()
```

3. List all libraries

Querying WRDS Data using Python

Learn how to use Python to view dataset metadata and query WRDS data

Introduction

Now that you've learned how to connect to WRDS using the Python **wrds** module, you're ready to begin querying WRDS data. This document will begin by examining the **wrds** module, presenting its included methods and covering the basics of its use. Afterwards, it will cover how to explore WRDS metadata to find the particular WRDS data that you're looking for, and then how to use that information to perform actual research queries against that data.

Additional Resources:

- [OLS Regression in Python](#)

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Using the wrds Module

All Python programs intending to access WRDS data must include the following two lines at the beginning:

```
1 import wrds
2 db = wrds.Connection()
```

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```
1 data = db.get_table('djones', 'djdaily', columns=['date', 'dji'], obs=10)
2 data
```

Using raw_sql()

The method `raw_sql()` accepts the following parameters:

- **sql** - the SQL string to query
- **date_cols** - a list or dict of column names to parse as date (optional)
- **index_col** - a string or list of column(s) to set as index(es) (optional)

For additional parameters, and further explanation of each, use the built-in help: `help(db.raw_sql)`

Here's an example that does the same thing as `get_table()` example above, but uses SQL to select the data instead:

```
1 data = db.raw_sql('SELECT date, dji FROM djones.djdaily LIMIT 10', date_cols=
2 data
```

The `raw_sql()` method is by far the most useful and popular of the **wrds** module's methods, allowing powerful and granular control over your data processing. Writing SQL for use with this method is fairly straightforward. All data queries are constructed in SQL the following generic manner:

```
select columns from library.dataset where column1 = value
```

Notice the *dot notation* for the library and dataset. Unlike the other **wrds** methods, where *library* and *table* are specified separately, SQL queries instead use the two together to identify the data location. So, for example, a data query for the *dataset* **msf** within the *library* **crsp** would use the syntax **crsp.msf**, and the same goes for **djones.djdaily**.

You'll likely be doing most of your work using the `raw_sql()` method.

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Related Information

[» PYTHON: From Your Computer](#)[» PYTHON: On the WRDS Cloud](#)[» Graphing Data](#)[» Example Workflow](#)

Access WRDS via Google Colab



```
import wrds
db = wrds.Connection()
```

... Enter your WRDS username [root]:hmlim
Enter your password:

WRDS 2FA by DUO: Complete Your Enrollment

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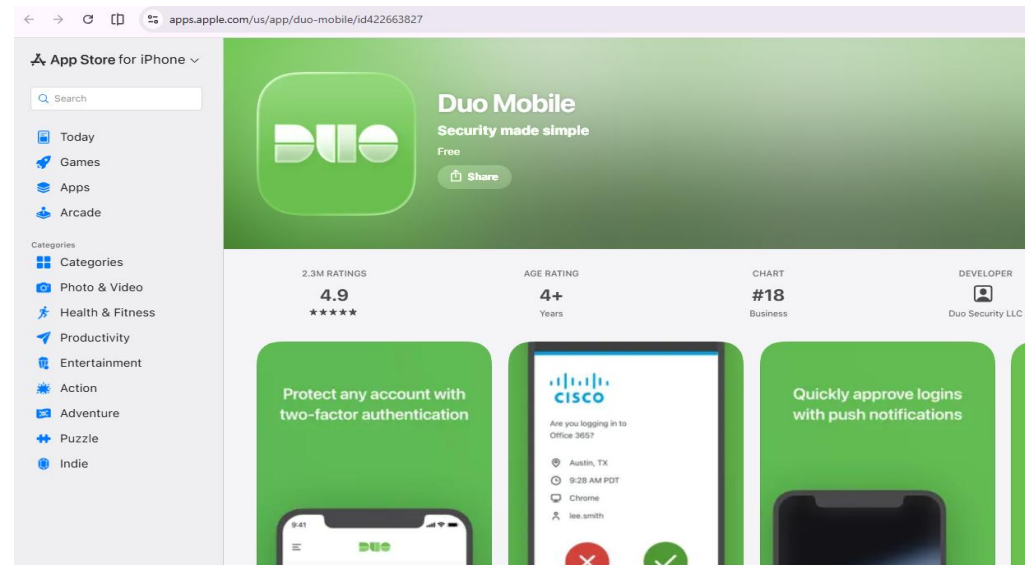
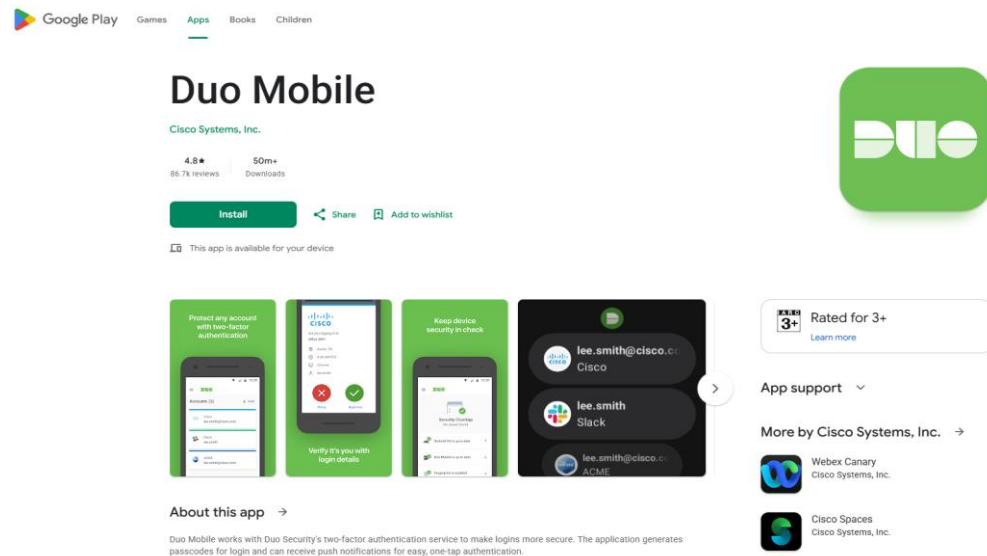
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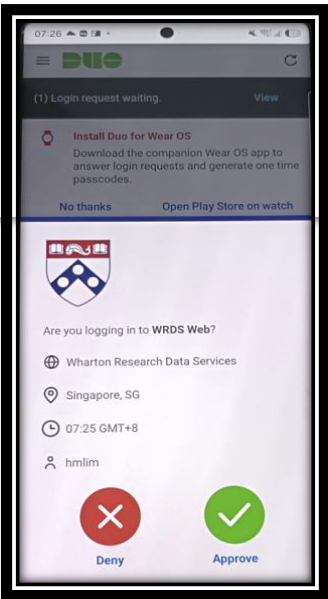
Best Regards,
The WRDS Team

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Philadelphia, PA 19104
United States

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Kind regards,
WRDS Support

Get table

```
dsf = db.get_table(library="crsp", table="dsf_v2", obs=10)
dsf.head()
```

	permno	hdrcusip	permco	siccd	nasdissuno	yyyymmdd	sharetype	securitytype	securitysubtype	usincflg	...	dlyopen	dlynumtrd	dlymmcnt	dlyprcvol	dlycumfacpr	dlycumfacshr	cusip	ticker	exchangeti
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3	10000	68391610	7952	3990	10396	19860110	NS	EQTY	COM	Y	...	<NA>	<NA>	10	21250.0	1.0	1.0	68391610	OMFGA	S
4	10000	68391610	7952	3990	10396	19860113	NS	EQTY	COM	Y	...	<NA>	<NA>	10	14306.3	1.0	1.0	68391610	OMFGA	S

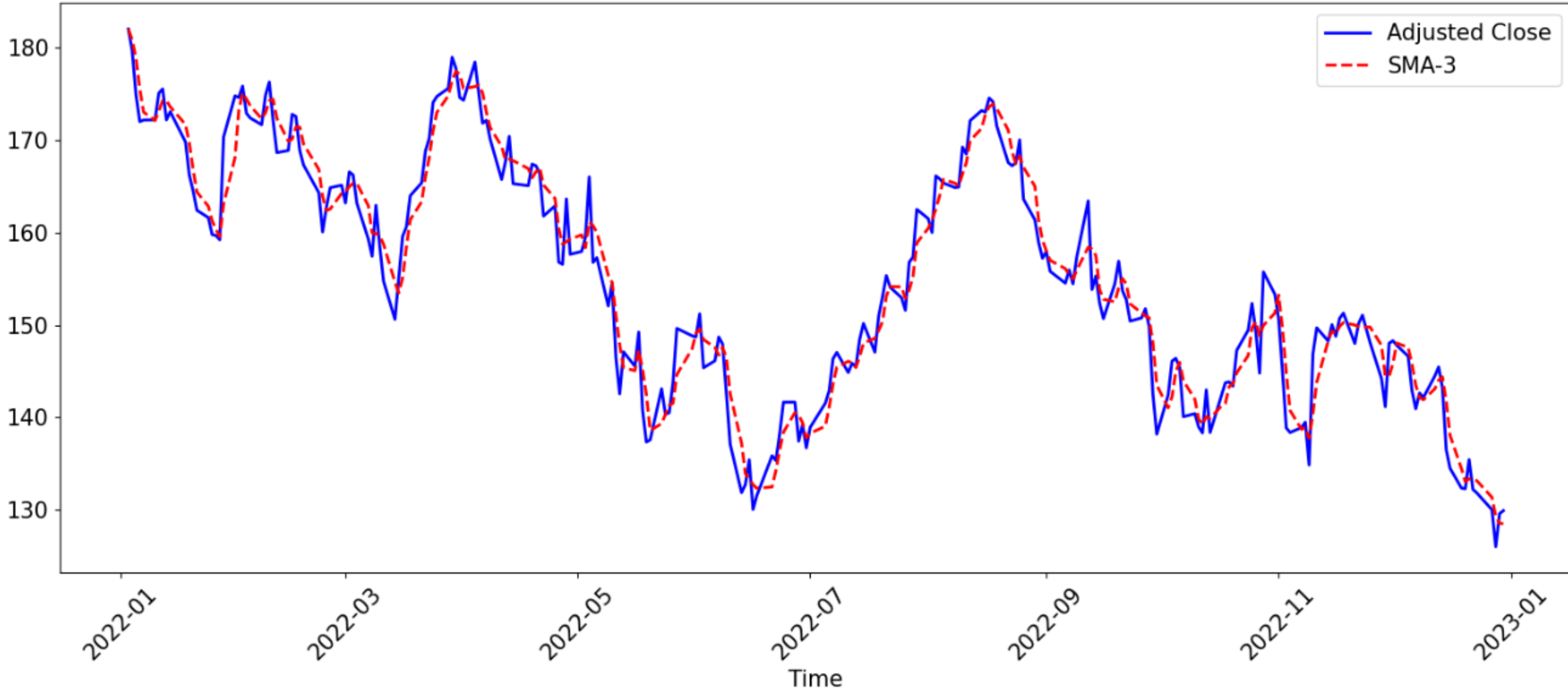
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Plot daily adjusted closing price



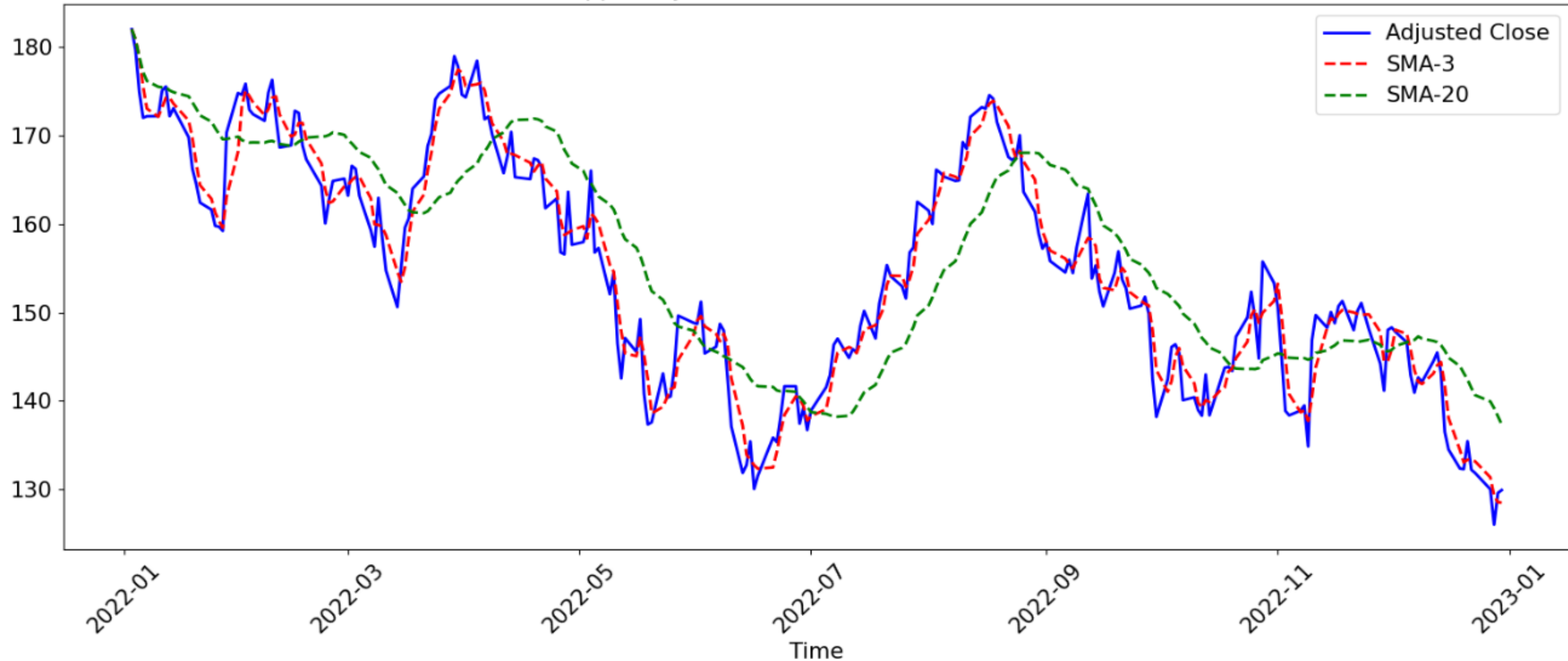
Plot daily adjusted closing price and simple moving average (SMA) 3 days

Apple Adjusted Close and SMA-3



Plot daily adjusted closing price and simple moving average 3 days and 20 days

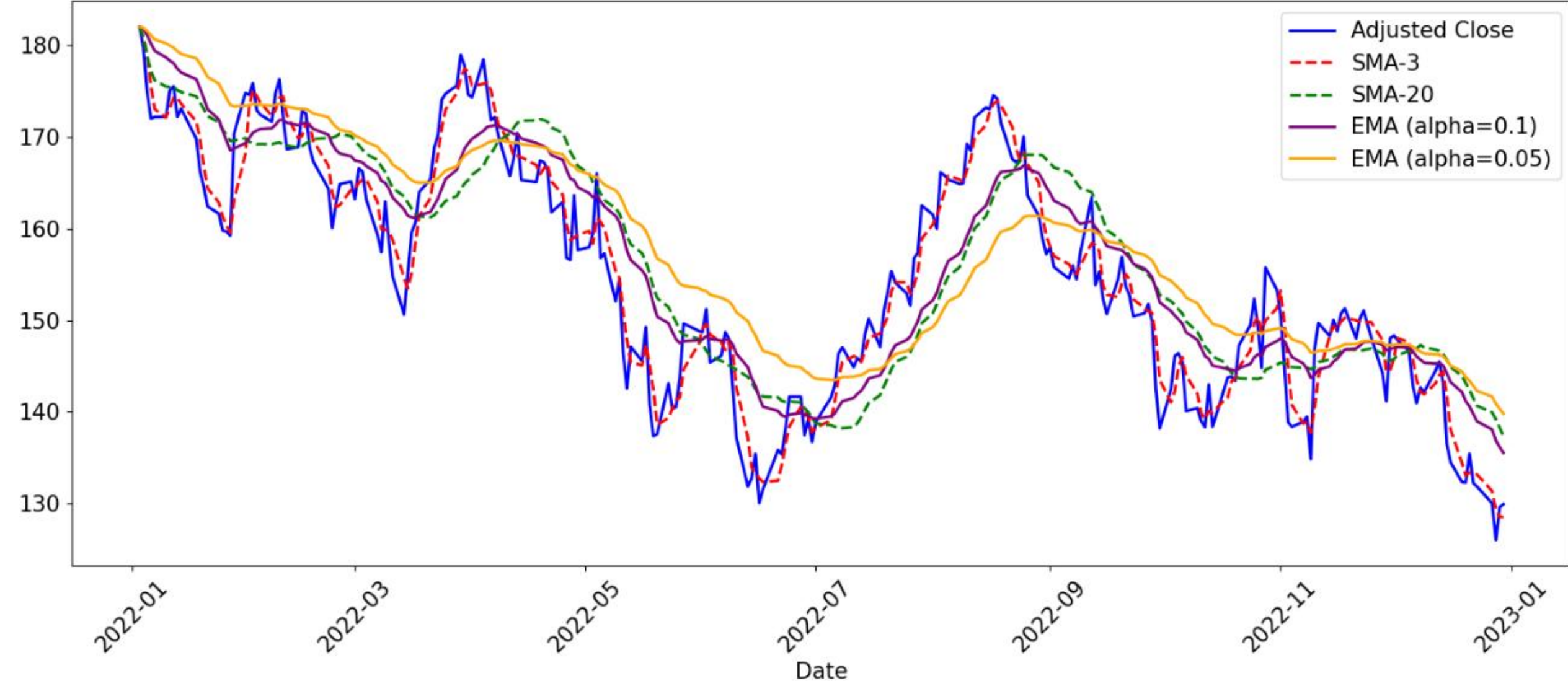
Apple Adjusted Close, SMA-3, and SMA-20



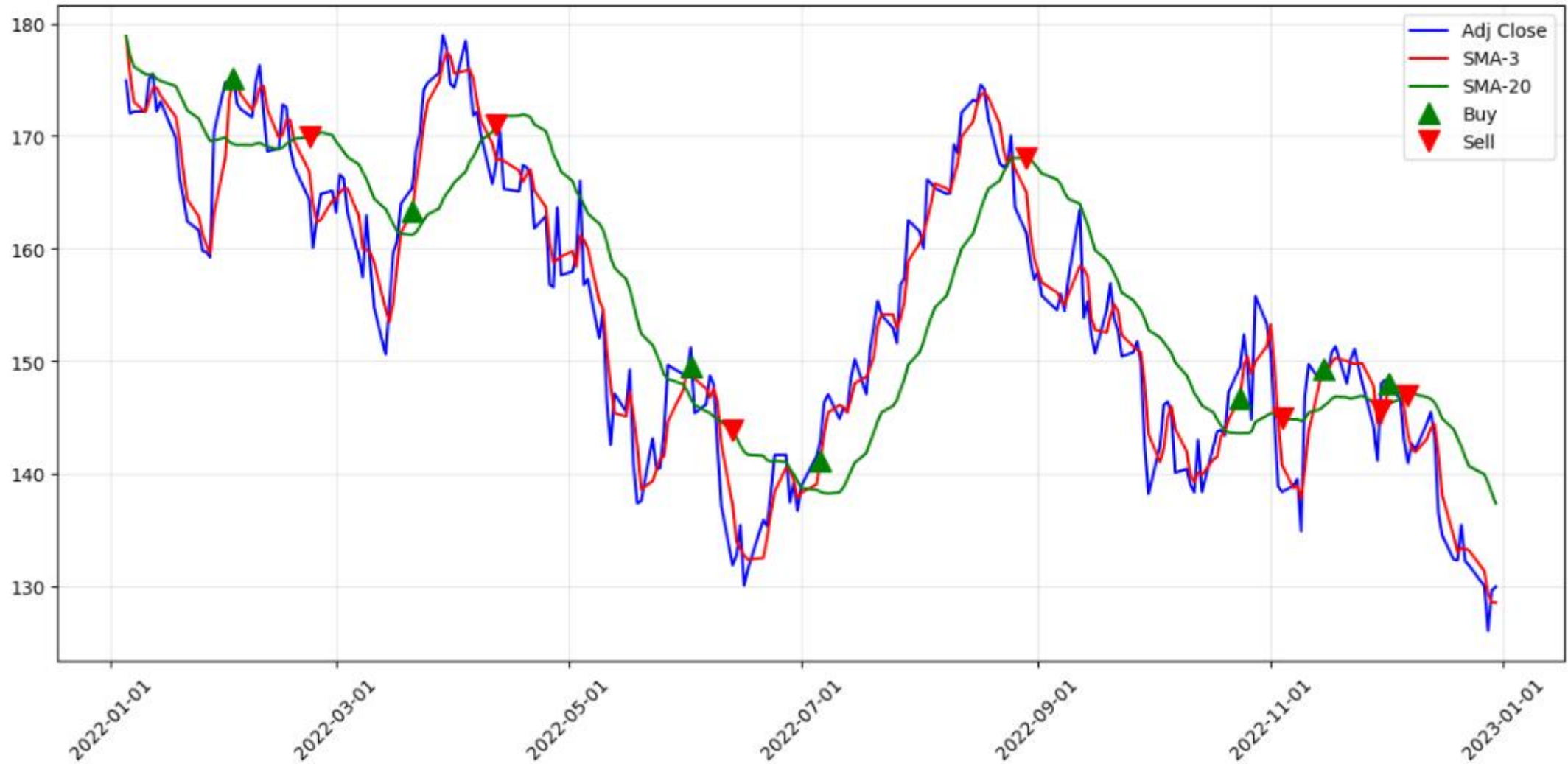
SMU Classification Restricted

Plot daily adjusted closing price and SMA 3 days, SMA 20 days and exponential moving average

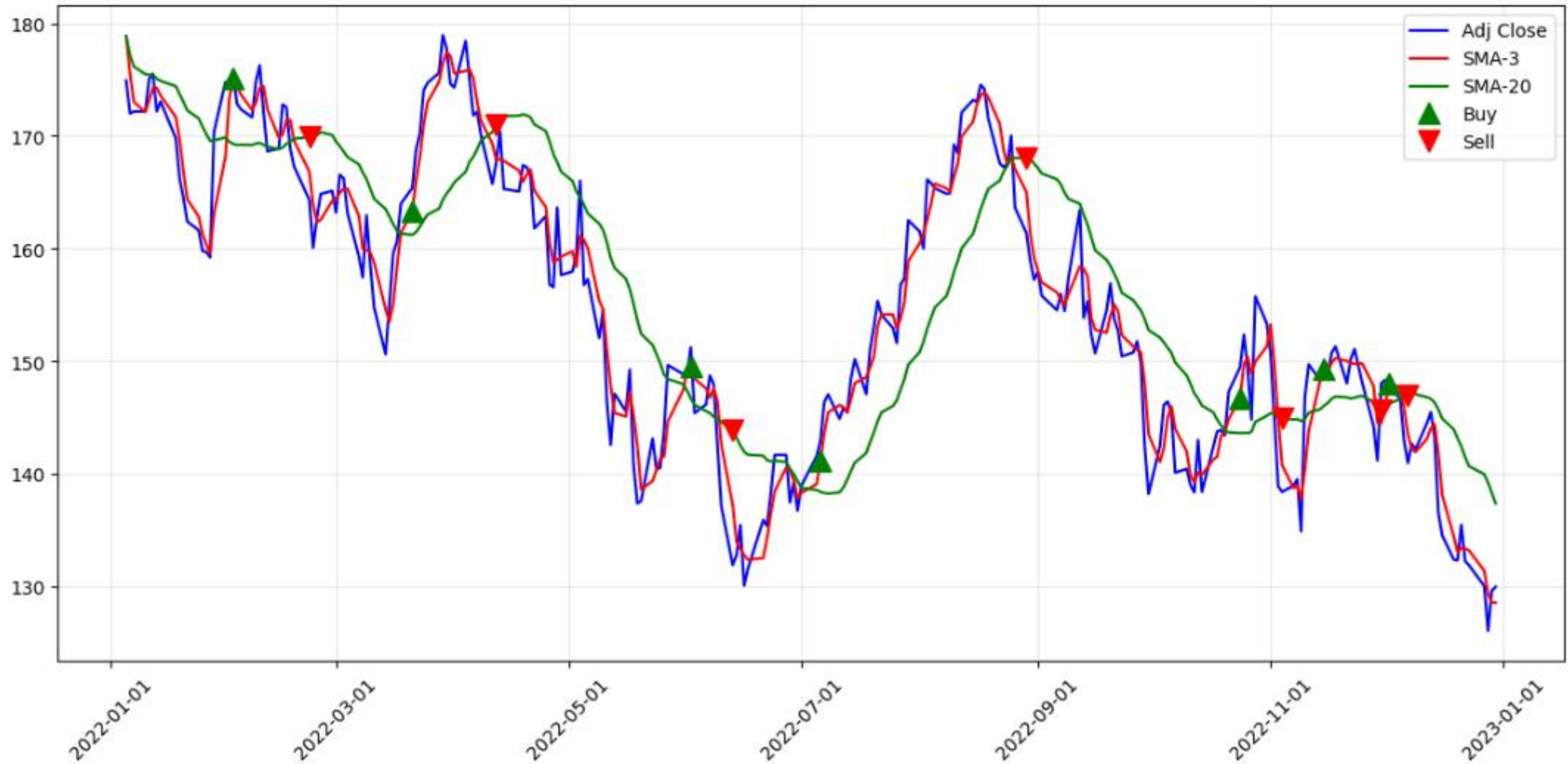
Daily Apple Adjusted Closing price with SMA-3, SMA-20, and EMA (alpha=0.1 and alpha =0.05)



Plot trading actions, from short to long (green triangles) and long to short (red triangles)



Plot trading actions, from short to long (green triangles) and long to short (red triangles)



WRDS

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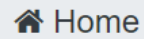
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WRDS News

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August 11th, 2026

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Description*

Please enter the details of your request. A member of our support staff will respond as soon as possible.

Paragraph | B | I |  |  |  |  | 99 |  | 

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